

المعطيات:

$$m(t) = 10 \sin(400\pi t) \text{ V} \Rightarrow f_m = 200 \text{ Hz}$$

$$x_c(t) = 50 \sin(2500\pi t) \Rightarrow f_c = 1250 \text{ Hz}$$

$$A_c = 50, \mu = 0.8$$

① Sideband Frequencies:

$$* f_{\text{USB}} = f_c + f_m = 1250 + 200 = 1450 \text{ Hz}$$

$$* f_{\text{LSB}} = f_c - f_m = 1250 - 200 = 1050 \text{ Hz}$$

② Amplitude of each sideband:

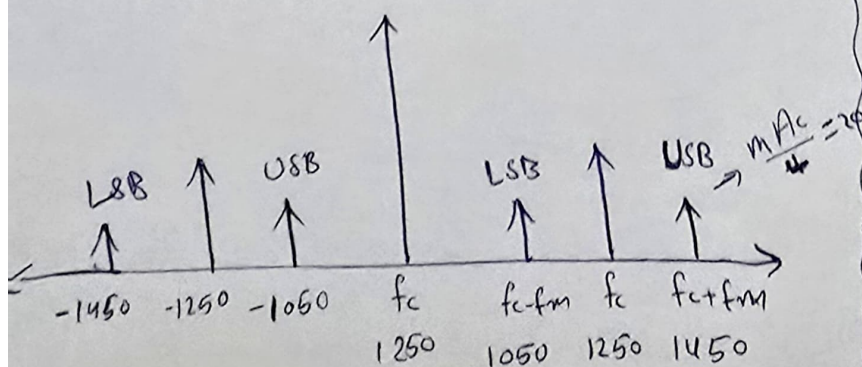
$$\frac{\mu A_c}{2} = \frac{0.8 \times 50}{2} = 20 \text{ V (each)}$$

③ AM efficiency (Power)

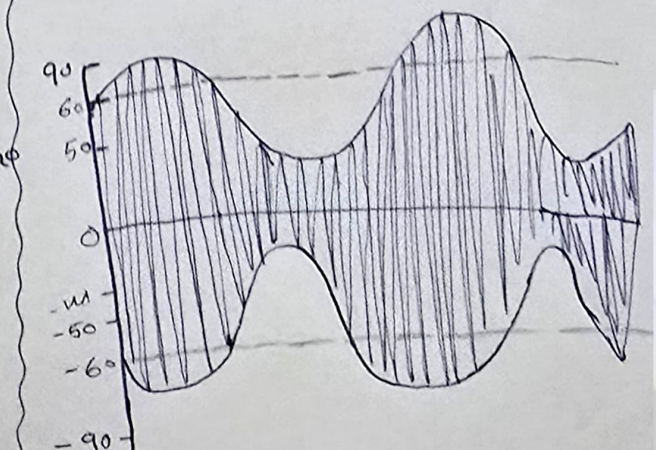
$$\eta = \frac{\mu^2}{2 + \mu^2} = \frac{(0.8)^2}{2 + (0.8)^2} = \frac{0.64}{2.64} \approx 0.2424 = 24.24\%$$

④

Frequency Spectrum



Modulated wave



المعطيات:

$$m(t) = 2 \cos(400\pi t) + 15 \sin(600\pi t) + \cos(1200\pi t)$$

① Component Frequencies:

$$\therefore f = \frac{\omega}{2\pi}$$

$$\therefore f_1 = \frac{400\pi}{2\pi} = 200 \text{ Hz}$$

$$f_2 = \frac{600\pi}{2\pi} = 300 \text{ Hz}$$

$$f_3 = \frac{1200\pi}{2\pi} = 600 \text{ Hz}$$

② Maximum Frequency:

$$\therefore 600 \text{ Hz} > 300 \text{ Hz} > 200 \text{ Hz} \quad \therefore f_{\max} = 600 \text{ Hz}$$

③ Nyquist rate and Interval:

$$\ast \text{ nyquist rate} \Rightarrow f_s = 2f_m = 2 \times 600 = 1200 \text{ Samples/s}$$

$$\ast \text{ nyquist Interval} \Rightarrow \frac{1}{f_s} = \frac{1}{2f_m} = \frac{1}{1200} \approx 0.000833 \text{ s} = 0.833 \text{ ms}$$

حل

$$x(t) = 3 \cos(100\pi t) + 10 \sin(600\pi t) + \cos(800\pi t)$$

$$f_{c1} = 300 \text{ Hz}, f_{c2} = 700 \text{ Hz}, f_{c3} = 1500 \text{ Hz}$$

$$f_{m1} = 50 \text{ Hz}, f_{m2} = 300 \text{ Hz}, f_{m3} = 400 \text{ Hz}$$

① Bandwidth of each Modulating Signal (channel)

$$BW_1 = 2f_{m1} = 100 \text{ Hz}, BW_2 = 600 \text{ Hz}, BW_3 = 800 \text{ Hz}$$

$$\text{channel 1 : } f_{c1} \pm f_{m1} \rightarrow 300 \pm 50 \rightarrow [250, 350]$$

$$\text{channel 2 : } f_{c2} \pm f_{m2} \rightarrow 700 \pm 300 \rightarrow [400, 1000]$$

$$\text{channel 3 : } f_{c3} \pm f_{m3} \rightarrow 1500 \pm 400 \rightarrow [1100, 1900]$$

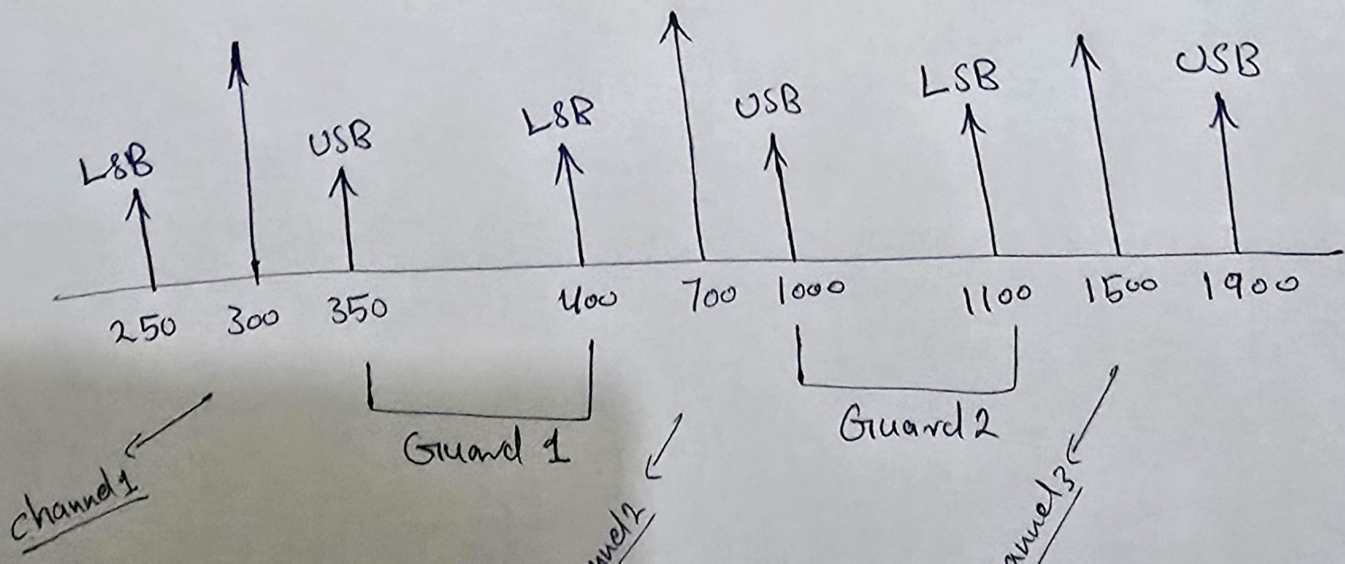
② Total Bandwidth and Guard band :

$$BW_t = 1900 - 250 = 1650 \text{ Hz}$$

$$\text{Guard 1 : } 400 - 350 = 50 \text{ Hz}$$

$$\text{Guard 2 : } 1100 - 1000 = 100 \text{ Hz}$$

③ Frequency Spectrum :



المعطيات

$$x(t) = 30 \cos(2\pi \times 10^6 t + 5 \sin(2\pi \times 2000 t))$$

$$\therefore A_c = 30V, f_c = 1MHz = 10^6 Hz, f_m = 2KHz, \beta = 5$$

① Frequency Deviation (Δf)

$$\Delta f = \beta f_m = 5 \times 2KHz = 10KHz$$

② Bandwidth:

$\because \beta > 1$, the Bandwidth $\rightarrow$ wide band

$$\therefore Bw = 2(\beta + 1)f_m = 2(5 + 1)2 = 24KHz$$

$$\text{or } Bw = 2(\Delta f + f_m) = 2(10 + 2) = 24KHz$$

③ Power FM wave:

$$P = \frac{A_c^2}{2R} = \frac{(30)^2}{2} = 450 \text{ watts}$$

المعطيات :

$$S(t) = 12 \cos(2\pi(1.2 \times 10^6)t + 7 \sin(2\pi \cdot 3000t)) \text{ volts}$$

$$A_c = 12V, f_c = 1.2 \text{ MHz}, f_m = 3 \text{ KHz}, \beta = 7$$

① Power in the modulated signal:

$$P = \frac{A_c^2}{2R} = \frac{12^2}{2} = 72 \text{ W}$$

② Peak Frequency Deviation (Δf)

$$\Delta f = \beta f_m = 7 \times 3 = 21 \text{ KHz}$$

③ Deviation Ratio

$$\rightarrow \frac{\Delta f}{f_m} = \beta = 7$$

④ Transmission Bandwidth:

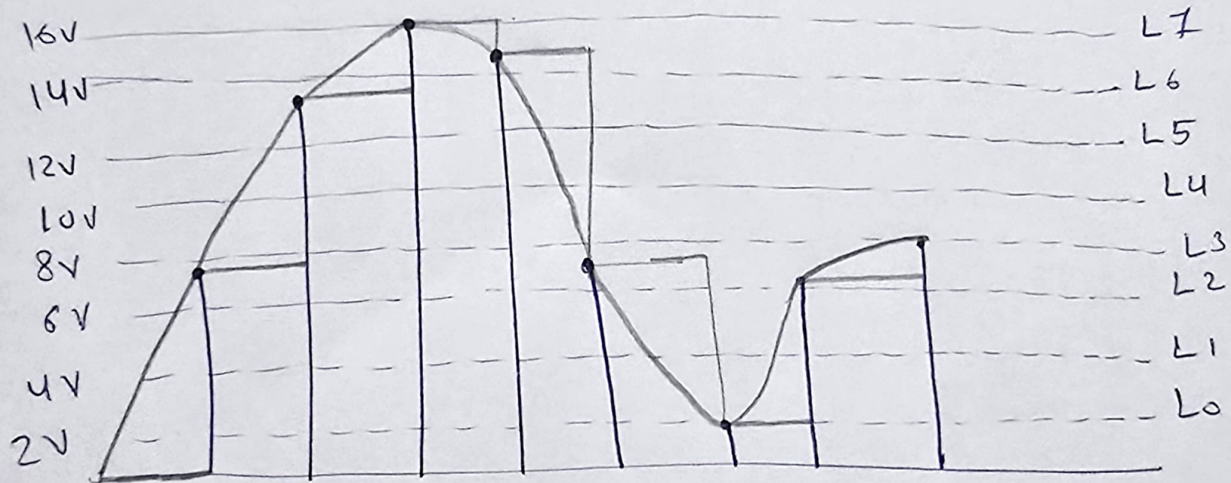
$\because \beta > 1 \therefore$ The BW is wide band

$$\therefore BW = 2(\beta + 1)f_m = 2(7 + 1)3 = 48 \text{ KHz}$$

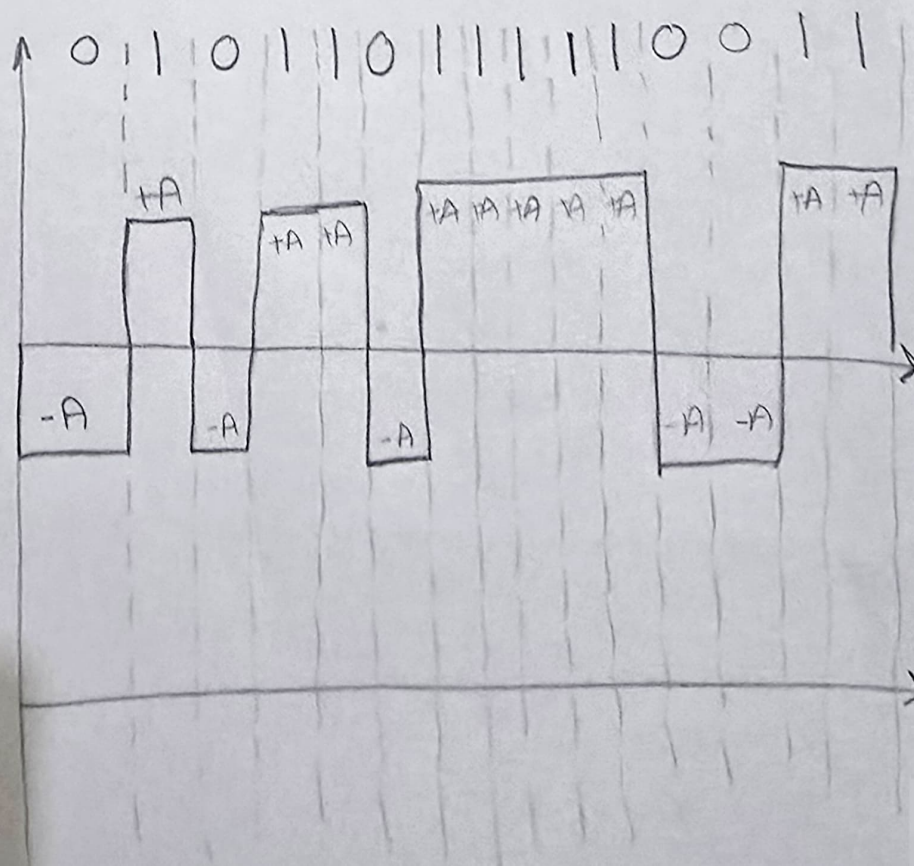
$$\text{or } = 2(\Delta f + f_m) = 2(21 \text{ KHz} + 3 \text{ KHz}) = 48 \text{ KHz}$$

Sampled ---	7.7	13.7	16v	14.3v	6.8v
Nearest Quantizing---	8v	14v	16v	14v	6v
Quantization error	-0.3	-0.3	0.0	0.3	0.8
Binary code	L3	L6	L7	L6	L2
Binary code	011	110	111	110	010

$\Rightarrow 010 \ 110 \ 111 \ 110 \ 011$



POLAR[NRZ]



POLAR(RZ)